National University of Computer and Emerging Sciences



LABORATORY MANUAL

for

Data Warehousing & Business Intelligence Lab

(DL3004)

LAB 02 • Power BI Fundamentals & Introductory BI Reporting

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Lab Manual 02

Objectives

- Import structured data into Power BI and inspect data types and quality in Power Query.
- Create and validate basic one-to-many relationships between a sales table and lookup tables.
- Build simple reusable measures for revenue, orders, quantity and average order value.
- Create management-oriented visuals: KPI cards, trend charts, category/region comparisons, a matrix and slicers.
- Configure visual interactions so filters behave intentionally rather than accidentally.
- Design a one-page dashboard that tells a coherent story and supports a management decision.

Tools	Power BI Desktop; supplied CSV files (PostgreSQL connection is optional)
Datasets	lab02_campuscart_sales.csv, products.csv, regions.csv
CLO Mapping	CLO 1
Submission	One .pbix file + exported dashboard/screenshot + insight summary

PRE-LAB: Continuity from Lab 1

The scenario is the same CampusCart business, but the data is now separated into Sales, Products and Regions so you can learn how Power BI relationships work. Formal dimensional modeling is taught later; today the focus is tool usage and reporting.

1. Business Scenario

CampusCart leadership wants a single-page BI dashboard that can answer “How are we performing, where, through which channels, and on which products?” Your task is to build the first operational BI report from three supplied tables.

DESIGN PRINCIPLE: Dashboard goal

A useful dashboard is not a collage of charts. It should let a manager start with overall KPIs, identify a trend, compare categories/regions, and then filter to investigate a problem.

2. Files & Intended Relationships

Sales	7,800 transaction rows; product_id and region_id act as lookup keys.
Products	One row per product; product name, category, brand and list price.
Regions	One row per region; region name, manager and service area.

Target relationship pattern

```
Products[product_id]  1  --------  *  Sales[product_id]
Regions[region_id]    1  --------  *  Sales[region_id]
```

Use single-direction filtering from the lookup table toward Sales.

COMMON PITFALL: Why uniqueness matters

The “1” side of a one-to-many relationship must contain unique keys. If Power BI refuses the relationship or shows many-to-many, check the key column for duplicates/nulls and confirm the correct table is on each side.

3. Import & Power Query Preparation

- Open Power BI Desktop > Get Data > Text/CSV and import all three CSV files.
- Choose Transform Data so that you inspect the data before loading it into the model.
- Rename queries to Sales, Products and Regions exactly; this keeps later instructions consistent.
- Check data types. IDs should be Whole Number/Text as appropriate, order_date should be Date, monetary fields Decimal Number, and discount_pct Whole Number.
- For text columns such as channel, customer_segment, category and region_name, inspect for unexpected blanks or inconsistent values.
- Apply changes and return to the report/model.

#	Task	Evidence
1	Confirm Sales row count and the min/max order_date.	Power Query evidence
2	Confirm Products[product_id] has no duplicates.	Validation evidence
3	Confirm Regions[region_id] has no duplicates.	Validation evidence

#	Task	Evidence
4	Set net_revenue_pkr and gross_revenue_pkr to Decimal Number.	Model evidence

4. Build the Data Model

- Switch to Model view.
- Create Products[product_id] -> Sales[product_id] as 1:*.
- Create Regions[region_id] -> Sales[region_id] as 1:*.
- Use single cross-filter direction.
- Do not create extra relationships “just because columns have similar names.” Every relationship should have a clear purpose.

CHECKPOINT: Model checkpoint

Your model should contain exactly three tables and two active one-to-many relationships. Take a screenshot before continuing.

5. Create Core Measures

5.1. Measure Setup

Create the following measures in the Sales table. These are deliberately basic; advanced DAX and time intelligence come later in the course.

Required measures

```
Total Net Revenue = SUM(Sales[net_revenue_pkr])

Total Orders = DISTINCTCOUNT(Sales[order_id])

Units Sold = SUM(Sales[quantity])

Average Order Value = DIVIDE([Total Net Revenue], [Total Orders])
```

- Format revenue measures as PKR/currency with sensible decimal places.
- Keep measure names business-friendly; avoid names such as Measure1 or SumOfRevenue.
- Verify that Total Orders changes when a slicer/filter is applied.

CONCEPT: Measure vs column

A measure is evaluated according to the current filter context. That is why the same Total Net Revenue measure can show a grand total in a card and category totals in a chart.

6. Build the Management Dashboard

6.1. Required Visuals

#	Task	Evidence
1	KPI cards: Total Net Revenue, Total Orders, Average Order Value.	3 cards

#	Task	Evidence
2	Monthly revenue trend: order_date by month on the axis, Total Net Revenue as value.	Line chart
3	Category performance: Products[category] vs Total Net Revenue, descending.	Bar chart
4	Regional performance: Regions[region_name] vs Total Net Revenue.	Column/bar chart
5	Region x Category matrix showing Total Net Revenue and Units Sold.	Matrix
6	Slicers for region_name, channel and customer_segment.	3 slicers

Minimum design requirements

- Use a clear report title and a subtitle stating the reporting period.
- Use meaningful chart titles (for example, “Net Revenue by Category” rather than “Bar Chart 1”).
- Avoid unnecessary 3D effects, decorative backgrounds and excessive colors.
- Keep KPI cards at the top, trend/comparison visuals in the middle, matrix/detail lower on the page.
- Make numbers readable at normal zoom and use consistent currency formatting.

6.2. Interactions & Testing

#	Task	Evidence
1	Select a region and confirm the KPI cards, trend and category visual update.	Interaction check
2	Select one category and confirm the matrix and regional visual respond appropriately.	Interaction check
3	Use Edit Interactions to disable at least one interaction that creates confusing behavior. Explain which interaction you disabled and why.	Screenshot + note
4	Clear all filters and confirm the dashboard returns to the full-year view.	Final check

QUALITY CHECK: Test like a user

Click every slicer and at least two chart elements before submission. A dashboard that looks correct but filters incorrectly is not complete.

7. Applied Challenge - The “Why?” Page

7.1. Investigate One Business Problem

Choose ONE of the following prompts. Add one additional visual that helps answer it, then write 3 concise insights underneath the report or in your submission document.

#	Task	Evidence
1	East region appears weaker than Central. Is the gap driven by product mix, channel mix, or fewer orders?	1 additional visual + insights
2	November has strong sales. Is the increase broad-based or concentrated in specific categories/channels?	1 additional visual + insights
3	Returns reduce net revenue. Which category/region combination should operations investigate first?	1 additional visual + insights

ASSESSMENT: Challenge standard

The additional visual must contribute new evidence. Duplicating an existing chart with a different color does not meet the challenge.

8. Submission Requirements

- Lab02_<RollNo>.pbix containing the final three-table model, measures and dashboard.
- One exported PDF/page screenshot of the dashboard in its unfiltered state.
- A screenshot of Model view showing both relationships.
- A short text/PDF containing your selected challenge prompt and 3 evidence-based insights.
- The report must open without missing data-source errors on the lab machine.

End of Lab Manual 02

Good Luck!!!